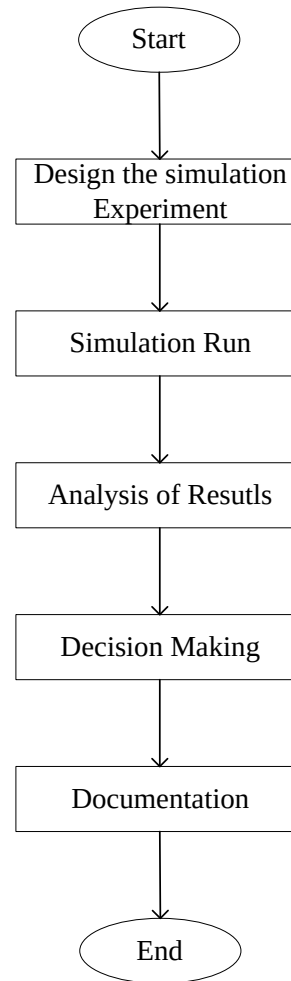


Simulation and Modeling

Lecture 12 **Simulation Execution**

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Simulation Program Execution (1/5)



Simulation Program Execution (2/5)

- Experiment Design
 - Initial condition design
 - Component of the model
 - In case of computer networks
 - Protocol stack
 - Protocol at different layer
 - No of nodes, routers
 - Source and destinations of the flows

Simulation Program Execution (3/5)

- Simulation Run
 - No of iterations
 - Management of results
 - For many runs, keeping raw data might not be necessary
 - Final data may be compiled for each run

Simulation Program Execution (4/5)

- Output Analysis
 - Statistical analysis of output data
 - Data of different simulation run might not be independent
 - They might be correlated
 - Unlike the assumption in classical statistics
 - Confidence interval
 - Extracting a detailed internal operation of the systems

- Decision Making
 - Based on the analysis of the output data
 - System behavior identified
 - Quantity of interest is analyzed
 - Final decisions are made
- Documentation
 - Specific observations and conjectures are documented
 - Explanation of the approximations
 - 1!1

Output Data Analysis

- Analysis of output data to draw conclusion
- A simulation is a computer based statistical sampling process
- Appropriate statistical techniques must be used
- Output of all simulations are non-stationary and auto-correlated
- Classical statistical tools not directly applicable
 - Because output data are not IID

Output data and IID

- Let $Y_1, Y_2, \dots$ is an output stochastic process from a single simulation run
- Y_i may represent the delay of the i -th job or throughput in the i -th hour
- Y_i 's are random variables – but not IID
- Classical statistical tools are applicable

Output data and IID (Cont'd)

- Let Y_{ij} are realization of Y_i in the j -th run
- Y_{ij} are found by using random numbers u_{ij}

$$y_{11}, \dots, y_{1i}, \dots, y_{1m}$$

$$y_{21}, \dots, y_{2i}, \dots, y_{2m}$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$y_{n1}, \dots, y_{ni}, \dots, y_{nm}$$

- Elements in the same row are from a single run and are not IID
- Elements in the same columns are the observation of the same job in different run and are IID
- This is the key to the output data analysis

Transient and Steady-state behavior

- $Y_1, Y_2, \dots$ is an output SP with initial condition I
- $F_{Y_i}(y_i|I) = P[Y_i \leq y_i|I]$ denote the conditional CDF of Y_i
- Usually $F_{Y_i}(y_i|I)$ are different
 - For each value of i , and
 - For each initial condition I
- Especially, for initial values of i , CDF will be different and might converge as $i \rightarrow \infty$

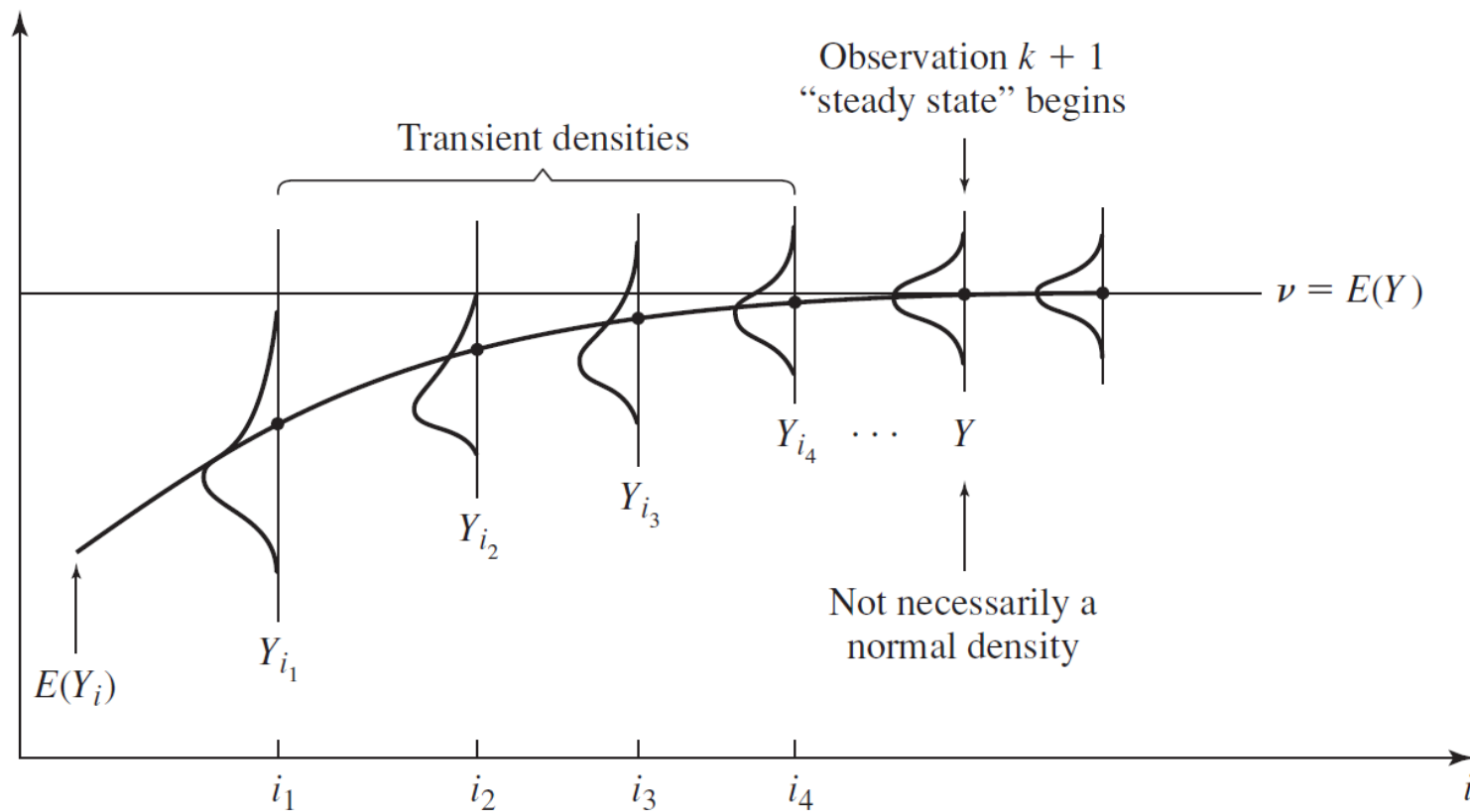


FIGURE 9.1

Transient and steady-state density functions for a particular stochastic process $Y_1, Y_2, \dots$ and initial conditions I .

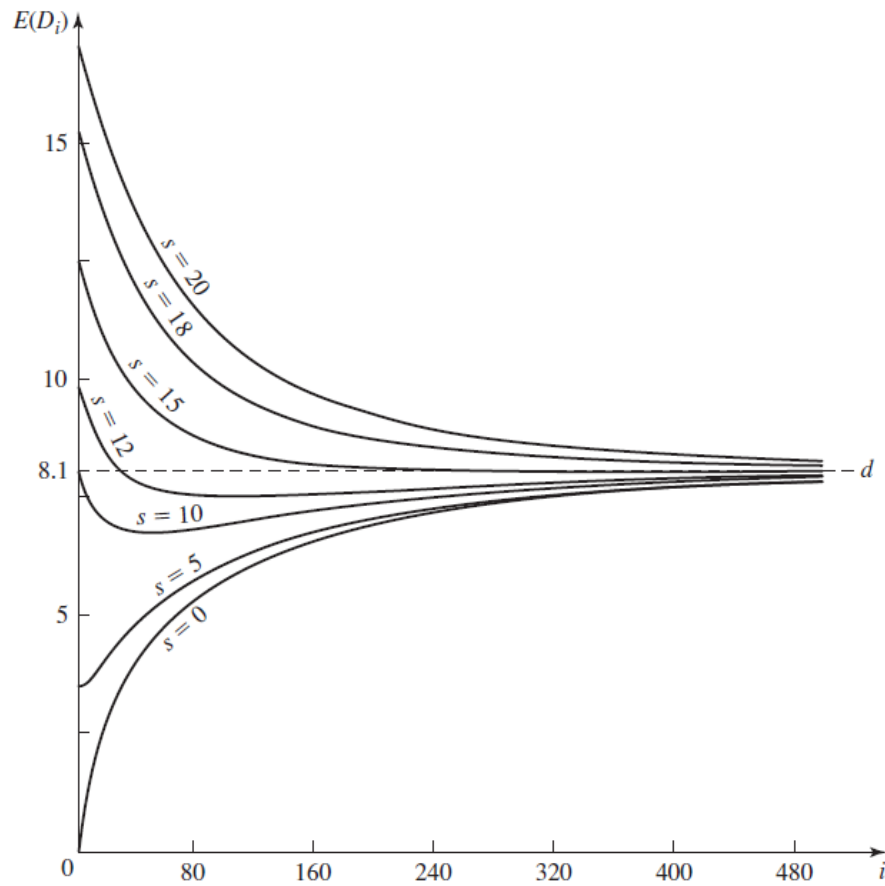


FIGURE 9.2

$E(D_i)$ as a function of i and the number in system at time 0, s , for the $M/M/1$ queue with $\rho = 0.9$.

Types of Simulation

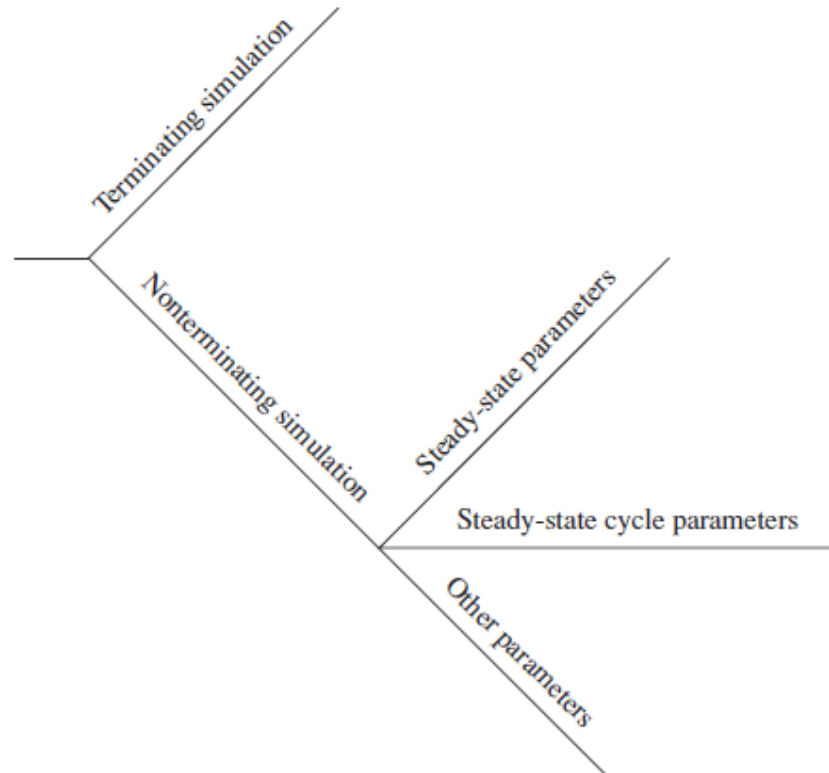


FIGURE 9.4

Types of simulations with regard to output analysis.

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- For terminating simulation, a natural event E exists that specifies the length of the simulation run
 - Initial conditions are important for terminating simulations
 - For non-terminating simulations, such an event does not exist to specify the length

Terminating

EXAMPLE 9.4. A retail/commercial establishment, e.g., a bank, closes each evening. If the establishment is open from 9 A.M. to 5 P.M., the objective of a simulation might be to estimate some measure of the quality of customer service over the period beginning at 9 A.M. and ending when the last customer who entered before the doors closed at 5 P.M. has been served. In this case $E = \{\text{at least 8 hours of simulated time have elapsed and the system is empty}\}$, and the initial conditions for the simulation are the number of customers present at time 0 (see Sec. 9.4.3).

EXAMPLE 9.5. Consider a military ground confrontation between a blue force and a red force. Relative to some initial force strengths, the goal of a simulation might be to determine the (final) force strengths when the battle ends. In this case $E = \{\text{either the blue force or the red force has “won” the battle}\}$. An example of a condition that would end the battle is one side losing 30 percent of its force, since this side would no longer be considered viable. The choice of initial conditions, e.g., the number of troops and tanks for each force, for the simulation is generally not a problem here, since they are specified by the military scenario under consideration.

Terminating

EXAMPLE 9.6. An aerospace manufacturer receives a contract to produce 100 airplanes, which must be delivered within 18 months. The company would like to simulate various manufacturing configurations to see which one can meet the delivery deadline at least cost. In this case $E = \{100 \text{ airplanes have been completed}\}$.

EXAMPLE 9.7. Consider a manufacturing company that operates 16 hours a day (two shifts) with work in process carrying over from one day to the next. Would this qualify as a terminating simulation with $E = \{16 \text{ hours of simulated time have elapsed}\}$? No, since this manufacturing operation is essentially a continuous process, with the ending conditions for one day being the initial conditions for the next day.

EXAMPLE 9.8. A company that sells a single product would like to decide how many items to have in inventory during a planning horizon of 120 months (see Sec. 1.5). Given some initial inventory level, the objective might be to determine how much to order each month so as to minimize the expected average cost per month of operating the inventory system. In this case $E = \{120 \text{ months have been simulated}\}$, and the simulation is initialized with the current inventory level.

Non-terminating

EXAMPLE 9.9. Consider a company that is going to build a new manufacturing system and would like to determine the long-run (steady-state) mean hourly throughput of their system after it has been running long enough for the workers to know their jobs and for mechanical difficulties to have been worked out. Assume that:

- (a) The system will operate 16 hours a day for 5 days a week.
- (b) There is negligible loss of production at the end of one shift or at the beginning of the next shift (see Prob. 9.3).
- (c) There are no breaks (e.g., lunch) that shut down production at specified times each day.

TABLE 9.1
Results for 10 independent replications of the bank model

Replication	Number served	Finish time (hours)	Average delay in queue (minutes)	Average queue length	Proportion of customers delayed < 5 minutes
1	484	8.12	1.53	1.52	0.917
2	475	8.14	1.66	1.62	0.916
3	484	8.19	1.24	1.23	0.952
4	483	8.03	2.34	2.34	0.822
5	455	8.03	2.00	1.89	0.840
6	461	8.32	1.69	1.56	0.866
7	451	8.09	2.69	2.50	0.783
8	486	8.19	2.86	2.83	0.782
9	502	8.15	1.70	1.74	0.873
10	475	8.24	2.60	2.50	0.779

Statistical Analysis

EXAMPLE 9.14. For the bank of Example 9.1, suppose that we want to obtain a point estimate and an approximate 90 percent confidence interval for the expected average delay of a customer over a day, which is given by

$$E(X) = E\left(\frac{\sum_{i=1}^N D_i}{N}\right)$$

(Note that we estimate the expected *average* delay, since each delay has, in general, a different mean.) From the 10 replications given in Table 9.1 we obtained

$$\bar{X}(10) = 2.03, \quad S^2(10) = 0.31$$

and

$$\bar{X}(10) \pm t_{9,0.95} \sqrt{\frac{S^2(10)}{10}} = 2.03 \pm 0.32$$

Thus, subject to the correct interpretation to be given to confidence intervals (see Sec. 4.5), we can claim with approximately 90 percent confidence that $E(X)$ is contained in the interval [1.71, 2.35] minutes.

EXAMPLE 9.15. For the inventory system of Sec. 1.5 and Example 9.8, suppose that we want to obtain a point estimate and an approximate 95 percent confidence interval for the expected average cost over the 120-month planning horizon, which is given by

$$E(X) = E\left(\frac{\sum_{i=1}^{120} C_i}{120}\right)$$

We made 10 independent replications and obtained the following X_j 's:

129.35	127.11	124.03	122.13	120.44
118.39	130.17	129.77	125.52	133.75

which resulted in

$$\bar{X}(10) = 126.07, \quad S^2(10) = 23.55$$

and the 95 percent confidence interval

$$126.07 \pm 3.47 \quad \text{or, alternatively,} \quad [122.60, 129.54]$$

Note that the estimated coefficient of variation (see Table 6.5), a measure of variability, is 0.04 for the inventory system and 0.27 for the bank model. Thus the X_j 's for the bank model are inherently more variable than those for the inventory system.

EXAMPLE 9.16. For the bank of Example 9.1, suppose that we would like to obtain a point estimate and an approximate 90 percent confidence interval for the expected proportion of customers with a delay less than 5 minutes over a day, which is given by

$$E(X) = E\left(\frac{\sum_{i=1}^N I_i(0, 5)}{N}\right)$$

where the *indicator function* $I_i(0, 5)$ is defined as

$$I_i(0, 5) = \begin{cases} 1 & \text{if } D_i < 5 \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, 2, \dots, N$. From the last column of Table 9.1, we obtained

$$\bar{X}(10) = 0.853, \quad S^2(10) = 0.004$$

and the 90 percent confidence interval

$$0.853 \pm 0.036 \quad \text{or} \quad [0.817, 0.889]$$

We first performed 500 independent simulation experiments for the $M/M/1$ queue with $\rho = 0.9$. For each experiment we considered $n = 5, 10, 20, 40$, and for each n we used (9.1) to construct an approximate 90 percent confidence interval for

$$d(25|s = 0) = E\left(\frac{\sum_{i=1}^{25} D_i}{25} \middle| s = 0\right) = 2.12$$

further discussion. The 90 percent confidence interval for the true coverage is computed from

$$\hat{p} \pm z_{0.95} \sqrt{\frac{\hat{p}(1 - \hat{p})}{500}}$$